economic environment it must now be retired and substituted by new technologies that do not burn fossil fuels.[10]

Technological breakthroughs can be helpful in many other fields besides energy. Consider, for example, the potential of developing ‘clean meat’. At present the meat industry not only inflicts untold misery on billions of sentient beings, but it is also one of the chief causes of global warming, one of the main consumers of antibiotics and poison, and one of the foremost polluters of air, land and water. According to a 2013 report by the Institution of Mechanical Engineers, it takes about 15,000 litres of fresh water to produce one kilogram of beef, compared to 287 litres needed to produce a kilogram of potatoes.[11]

The pressure on the environment is likely to get worse as rising prosperity in countries such as China and Brazil allows hundreds of millions of additional people to switch from eating potatoes to eating beef on a regular basis. It would be difficult to convince the Chinese and the Brazilians – not to mention the Americans and the Germans – to stop eating steaks, hamburgers and sausages. But what if engineers could find a way to grow meat from cells? If you want a hamburger, just grow a hamburger, instead of raising and slaughtering an entire cow (and transporting the carcass thousands of kilometres).

This might sound like science fiction, but the world’s first clean hamburger was grown from cells – and then eaten – in 2013. It cost $330,000. Four years of research and development brought the price down to $11 per unit, and within another decade industrially produced clean meat is expected to be cheaper than slaughtered meat. This technological development could save billions of animals from a life of abject misery, could help feed billions of malnourished humans, and could simultaneously help to prevent ecological meltdown.[12]

Hence there are many things that governments, corporations and individuals can do to avoid climate change. But to be effective, they must be done on a global level. When it comes to climate, countries are just not sovereign. They are at the mercy of actions taken by people on the other side of the planet. The Republic of Kiribati – an islands nation in the Pacific Ocean – could reduce its greenhouse gas emissions to zero and nevertheless be submerged under the rising waves if other countries don’t follow suit. Chad could put a solar panel on every roof in the country and yet become a barren desert due to the irresponsible environmental policies of distant foreigners. Even powerful nations such as China and Japan are not ecologically sovereign. To protect Shanghai, Hong Kong and Tokyo from destructive floods and typhoons, the Chinese and Japanese will have to convince the Russian and American governments to abandon their ‘business as usual’ approach.